

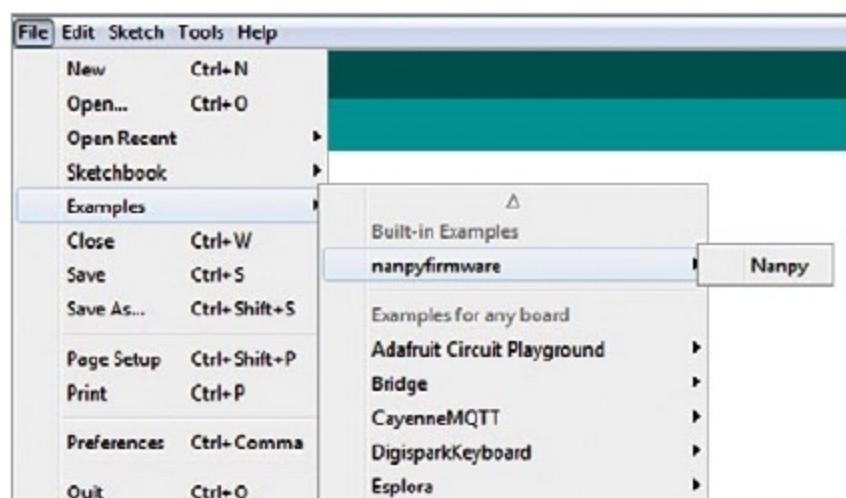


Fig 3: Arduino IDE window opens from Raspberry Pi OS

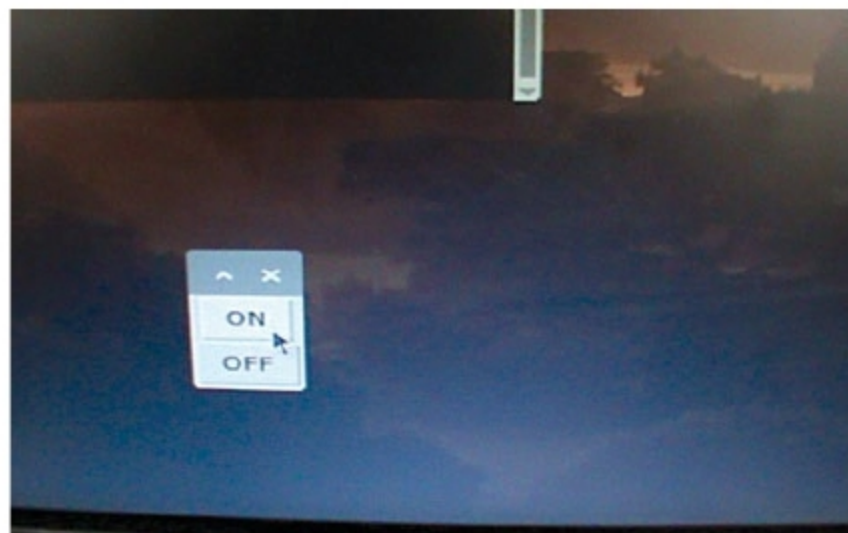


Fig. 4: GUI program with on and off controls

```
$ git clone https://github.com/nanpy/nanpy
```

6. Enter the following command:

```
$ git clone https://github.com/nanpy/nanpy-firmware
```

This will install nanpy firmware from Github in Downloads folder.

At this stage if you use ls command, there will be two folders listed: nanpy and nanpy-firmware. Rename nanpy-firmware as nanpyfirmware.

7. Enter the following command, followed by ls:

```
$ cd nanpyfirmware
```

This will take you to nanpyfirmware directory. List the contents with a file called configure.sh. Run this file by running the command given below.

```
$ ./configure.sh
```

8. Navigate to nanpy directory using the following command, followed by ls:

```
$ cd nanpy
```

It will display the directory con-

tents. nanpy directory will show a file called cfg.h. This file has the option to add Arduino library options that are needed by setting the contents to 1. In Raspberry Pi, contents of the file can be edited using nano or Leafpad. For example, to enable use of servo with nanpy, define the parameter as given below.

```
#define USE_Servo
(0=Disabled,1=Enabled.)
```

Likewise, to use One Wire device, define the parameter as given below.

```
#define USE_Wire
```

9. Copy nanpy files to Arduino sketchbook. But first open Arduino IDE on Raspberry Pi and make sure that Arduino IDE opens without any

errors. Then using File Manager make sure that there is /sketchbook/libraries directory in Home folder.

10. Navigate back to Downloads directory using cd.. command twice. Then, use the following command:

```
$ cp avr-nanpyfirmware/ ~/sketchbook/libraries
```

In some cases, this command will generate an error, so it may be better to copy the files to Home folder and then copy them to sketchbook/libraries.

Installation of Nanpy on Arduino IDE

The steps for installing Nanpy on Arduino IDE are:

1. Connect Arduino Uno board to Raspberry Pi using a standard USB cable. Open Arduino IDE and navigate to nanpy from Menu on the IDE.

Files→Sketchbook→libraries→nanpyfirmware→nanpy

2. Open Nanpy code from Arduino IDE (Fig. 3) and select Arduino Uno under Tools and the correct COM port. Compile and upload the code to Arduino board.

3. Go back to Linux terminal and navigate to Downloads folder and enter cd nanpy command, followed

by ls command. When ls command is executed, files in nanpy directory will be displayed and there will be a file called setup.py.

4. To install nanpy files for Python 3 on Raspberry Pi, use the following command:

```
$ sudo python3 setup.py install
```

This will install nanpy files for Python 3 on Raspberry Pi.

5. If Python 2 is used on Raspberry Pi, use the command given below.

```
$ sudo python setup.py install
```

After opening nanpy file, download the same to Arduino after verification. Arduino board is now ready as a slave to Raspberry Pi board. It will accept commands from Raspberry Pi and serve as requested.

Python programming and Tkinter to control Arduino I/O pin

In this section we will develop a Python application using Tkinter interface. In the latest operating system for Raspberry Pi, Python 3 and Tkinter come pre-installed.

As an example, we will write a Python code (Nanpy-Arduino_Slave.py) to establish a connection between Raspberry Pi and Arduino. Using two buttons in Tkinter, we can turn on and off the onboard LED connected to pin 13 on Arduino Uno.

Download Python code (Nanpy-Arduino_Slave.py) and open it on the terminal using the following commands:

```
$ cd Downloads
$ python3 nanpy-Arduino_Slave.py
```

The program first imports all required modules from nanpy and Tkinter. Then, using Tkinter, a frame is created along with two buttons—on and off.

Now, a GUI will open having two buttons (on and off). When you click on 'on,' the onboard LED will glow; when you click on 'off,' the LED will be off. The procedure is simple; just write 1 for on and 0 for off to pin 13 of Arduino. The final program output screenshot is shown in Fig. 4. **EFY**



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EFY Note
The source code of this project is available for free download at source.efymag.com